

Design Evaluation 1—XR Podcast Explorer

1. Objective and Validation Metrics

The objective of this evaluation was to determine whether first-time users could understand and complete the main spatial interaction flow of the **XR Podcast Explorer** without detailed instructions. Five participants completed a moderated task-based usability test covering the journey from **Podcast Hub** → **Episode Room** → **Chapter Portal** → **Podcast Experience**, followed by Favourite and return-navigation tasks.

Success was evaluated through **task completion, hesitation, errors, assistance required, direct observation and think-aloud feedback**. An interaction was considered successful when participants could understand its purpose and complete it independently with minimal hesitation. Repeated hesitation or moderator assistance was treated as evidence that further iteration was required.

Interaction tested	Validation criterion
Topic Door	Recognise the door as an entry point and enter the AI topic
Episode	Browse available Episodes and select one
Chapter	Browse Chapters, select Interview and enter it
Favourite	Understand and complete the Heart interaction independently
Navigation	Return to Episode Room and Podcast Hub
Spatial flow	Understand Topic → Episode → Chapter → Experience hierarchy

2. Results

Testing showed that the **core podcast exploration flow was generally understandable**, but interaction performance varied. Door-based entry and Episode browsing were the strongest interactions. The Favourite interaction produced the most repeated difficulty, while Chapter transitions and global navigation showed some uncertainty.

Area	Result	Key finding
Topic Door	Strong	Doors were generally understood as spatial entry points and were positively received.
Episode Browsing	Strong	Participants could browse and select Episodes; button-based controls were considered clear.
Chapter Selection	Mixed	Tasks were completed, but Chapter meaning and the Chapter → Current Chapter Door transition caused uncertainty.
Favourite	Weak	Multiple participants needed guidance to manipulate the Heart, despite generally recognising its favourite meaning.
Navigation	Mixed	Return navigation was understood overall, but panel behaviour and spatial relationships caused uncertainty.
XR Experience	Mixed	Doors supported immersion, but participants also described the environment as static or insufficiently spatial.

Recurring feedback included requests for clearer interaction cues, a more physical Heart, stronger spatial relationships between rooms, richer environmental content and improvements to supporting UI.

3. Analysis / Insights

Insight 1 — Spatial doors are the strongest part of the XR interaction

Doors consistently communicated the idea of **entering podcast content**. Participants could connect the familiar physical action of entering a door with navigating between digital content spaces.

This validates the use of a spatial metaphor rather than relying entirely on conventional menus. **Topic Door and Current Chapter Door interactions should therefore be retained and strengthened.**

Insight 2 — The Heart metaphor works, but its affordance does not

Participants often understood that the Heart represented **favouriting or collecting content**, yet several did not understand that they needed to physically move it to the Favourite Area.

This reveals a distinction between **semantic clarity and interaction clarity**: the concept is understandable, but the required action is not sufficiently discoverable. The Heart metaphor should therefore remain, while its size, three-dimensionality, grab feedback and placement feedback should be improved.

Insight 3 — Logical hierarchy is clearer than spatial hierarchy

Most participants understood the progression between Topic, Episode, Chapter and Podcast Experience.

However, uncertainty occurred at some transitions, particularly between Chapter selection and the Current Chapter Door. One participant also described room-to-room teleportation as abrupt and suggested a spatial map.

Therefore, users generally understand **what level they are navigating**, but not always **where spaces are in relation to each other**. Stronger current-location and transition cues are needed.

Insight 4 — The concept is spatial, but parts of the experience remain too 2D/static

Participants responded positively to embodied interactions such as doors, while conventional panels and limited environmental movement were perceived as less immersive. Some participants specifically wanted greater movement and richer environmental content.

The next iteration should therefore prioritise **physical and spatial interaction**, while using flat UI only where it clearly supports usability.

4. Evaluation of Aims

The overall testing objective was **partially validated**. First-time users generally understood the XR podcast concept and completed the main exploration flow, but several interactions still required assistance.

Testing Aim	Evaluation	Evidence
Understand Topic Doors	Validated	Door entry was consistently understandable and positively received.
Browse/select Episodes	Validated	Episode browsing and selection were generally completed successfully.
Understand Chapter navigation	Partially Validated	Chapter meaning and the transition to the Current Chapter Door caused uncertainty.
Independently complete Favourite	Invalidated	Multiple participants required assistance with the Heart interaction.
Understand return navigation	Partially Validated	Purpose was understood, but panel behaviour and spatial context caused difficulties.

Testing Aim	Evaluation	Evidence
Understand spatial hierarchy	Partially Validated	Logical flow was clear, but some transitions felt abrupt or insufficiently spatial.

Overall, testing **validated the core spatial podcast concept and door metaphor**, while identifying Favourite affordance, Chapter transition cues and spatial continuity as the main areas requiring iteration.

5. Concept Iteration

Evaluation findings were translated into design changes according to their impact on independent task completion.

Insight / Problem	Design Response	Priority
Heart action is unclear	Make Heart larger and clearly 3D ; improve grab affordance and feedback	High
Favourite result is unclear	Keep the placed Heart visibly attached to the Favourite Area	High
Chapter → Door transition is unclear	Highlight Current Chapter Door after selection and add an “Enter selected chapter” cue	High
Spatial relationships are weak	Explore a spatial/map-based Navigation representation	Medium
Experience feels static	Increase opportunities for spatial movement and environmental exploration	Medium
Supporting UI needs refinement	Improve Navigation layout, Episode presentation and subtitle controls	Low–Medium

The highest-priority changes directly address interactions that prevented independent task completion.

Environmental richness and supporting UI improvements remain important but will follow the core usability fixes.

The successful **door-based interaction will be retained**, demonstrating that evaluation does not only identify features to remove or fix; it also provides evidence for design decisions that should be affirmed.

6. Reflection and Next Steps

This evaluation demonstrated that **task completion alone does not indicate successful usability**. Some participants eventually completed tasks but required hesitation, clarification or assistance. Observing how users approached each interaction was therefore more informative than recording success alone.

The testing also revealed a difference between designing an interaction that appears understandable during development and making it discoverable to a first-time user. The Heart was the clearest example: its meaning was recognisable, but its physical action was not.

A limitation of this round was that some sessions were affected by prototype setup/state issues, including Chapter labelling and Navigation availability. A consistent **pre-test reset checklist** will be used in the next evaluation so that design issues can be more clearly separated from implementation problems.

The next prototype will prioritise **Heart affordance, Chapter-to-door transition cues, navigation clarity and spatial movement**. Evaluation 2 will test whether these changes reduce hesitation and assistance while preserving the successful door interaction. The goal is to move from a prototype whose overall structure is understandable to an XR experience that users can navigate **independently, spatially and confidently**.

Appendix: Testing raw data

IP1 Testing Result – Participant P1

Participant ID	P1
Prototype	Interactive Prototype 1

Task Performance

Task	Success	Errors / Issues	Hesitation	Assistance	Key observation
T1 Topic Selection	Yes	No clear issue observed	No	No	The participant understood that the Topic Doors represented entry points and successfully selected the AI podcast topic.
T2 Episode Browsing & Selection	Yes	No major issue observed	Low	No	The participant understood the Episode browsing and selection process and was able to choose an Episode of interest.
T3 Chapter Browsing & Selection	Yes, with assistance	Transition from Chapter selection to the Current Chapter Door was unclear	Yes	Yes	After selecting a Chapter, the participant was unsure whether to use another interface control or interact with the door. A clearer cue is needed to show that the Current Chapter Door is the next step.
T4 Favourite Interaction	Partially completed	Heart was difficult to grab and placement feedback was unclear	Yes	Yes	The participant understood the Heart as a favourite mechanism and liked the concept, but found it difficult to pick up. The placed Heart should remain clearly visible on the Favourite Area surface.
T5 Global Navigation	Yes, with assistance	Navigation Panel had no obvious close/back action	Yes	Yes	The participant understood the purpose of Navigation but was unsure how to dismiss the panel and required prompting during the return flow.

Task-Specific Observations

Observation	Result
Recognised Topic Door as an entry interaction	Yes
Understood Episode browsing/selection concept	Yes
Understood Chapter selection → Current Chapter Door relationship	No
Understood Heart as a favourite interaction	Yes

Could reliably grab the Heart	No
Preferred Heart to remain visibly attached to Favourite Area	Yes
Understood Navigation as a return/navigation control	Yes
Found an obvious close/back action for Navigation Panel	No
Expected podcast audio playback	Yes

Post-Test Interview Notes

Question	Participant response / notes
Q1. Which interaction was the easiest to understand, and why?	The participant found the Topic Door interaction easy to understand because the door clearly suggested an entry point. The Heart-based favourite concept was also intuitive and was one of the participant's preferred interactions.
Q2. Was there any point where the participant was unsure about what to do next?	The main uncertainty occurred after selecting a Chapter. The participant did not know whether to use another interface control or interact with the Current Chapter Door. The participant was also unsure how to close or go back from the Navigation Panel.
Q3. What was the participant's initial interpretation of the Heart and Favourite Area?	The participant understood that the Heart represented favouriting and liked the concept. However, the Heart was difficult to pick up. The participant suggested keeping the Heart visibly attached to the Favourite Area surface so the result of the action is easier to recognise.
Q4. Did the Topic → Episode → Chapter → Podcast Experience flow feel clear? Why or why not?	The overall spatial flow was generally understandable, particularly the use of doors as entry points. The main breakdown occurred between Chapter selection and entering the Podcast Experience because the prototype did not clearly indicate that the selected Chapter should be entered through the Current Chapter Door.
Q5. If one thing could be changed about the experience, what would the participant change?	The participant prioritised a clearer visual cue for entering the Current Chapter Door, a close/back action for the Navigation Panel, and a more reliable Heart interaction with clearer placement feedback.

Summary Notes

Additional observations:

Door-based navigation and the Heart favourite metaphor were relatively intuitive. The main difficulties were caused by missing interaction cues and prototype limitations. The participant also expected podcast audio playback.

Main usability issues identified:

1. The transition from Chapter selection to the Current Chapter Door is not sufficiently signposted.
2. The Navigation Panel does not provide an obvious close/back action.
3. The Heart interaction is difficult to grab and its placement feedback is visually unclear.

Potential design improvements:

1. Add a clear text or icon cue to the Current Chapter Door, such as "Enter selected chapter".
2. Add a close/back control to the Navigation Panel.
3. Improve the Heart grab interaction and keep the placed Heart clearly visible on the Favourite Area surface.
4. Add subtle affordance cues for interactive doors.

5. Add podcast audio playback in a later prototype iteration.

IP1 Testing Result – Participant P2

Participant ID	P2
Prototype	Interactive Prototype 1

Task Performance

Task	Success	Errors / Issues	Hesitation	Assistance	Key observation
T1 Topic Selection	Yes	No clear issue observed	No	No	The participant quickly understood the Topic Door interaction and successfully entered the AI podcast topic.
T2 Episode Browsing & Selection	Yes	No clear issue observed	No	No	The participant found the Episode browsing and selection interaction easy and straightforward.
T3 Chapter Browsing & Selection	Yes, with clarification	Incorrect Chapter label affected the task	Yes	Yes	An incorrect Chapter label created uncertainty during selection. The participant required clarification to identify the intended Interview chapter.
T4 Favourite Interaction	Yes	No major issue observed	No	No	The participant successfully completed the Favourite interaction.
T5 Global Navigation	Yes, with assistance	Navigation Panel did not appear as expected during the return workflow	Yes	Yes	The expected Navigation Panel was unavailable during the return workflow, so guidance was required to complete the return to the Podcast Hub.

Task-Specific Observations

Observation	Result
Recognised Topic Door interaction	Yes
Understood Episode browsing/selection controls	Yes
Browsed available Episodes before selection	Yes
Browsed available Chapters before selection	Yes
Chapter task affected by incorrect labelling	Yes
Completed Heart favourite interaction	Yes
Found Navigation without assistance	No
Returned to Episode Room / Podcast Hub	Yes, with assistance
Liked the physical door-opening interaction	Yes

Post-Test Interview Notes

Question	Participant response / notes
Q1. Which interaction was the easiest to understand, and why?	The participant found the Topic Door interaction and Episode browsing/selection easy to understand. The door interaction was especially positive because the opening motion created a satisfying physical push-open feeling.
Q2. Was there any point where the participant was unsure about what to do next?	Uncertainty occurred during the Chapter task because an incorrect label affected the available choice. A second difficulty occurred during the return-navigation task when the expected Navigation Panel did not appear.
Q3. What did the participant think about the Favourite interaction?	The participant successfully completed the Favourite interaction and did not identify it as a major usability problem.
Q4. Did the Topic → Episode → Chapter → Podcast Experience flow feel clear? Why or why not?	The participant considered the overall movement between Topic, Episode, Chapter and Podcast Experience clear.
Q5. If one thing could be changed about the experience, what would the participant change?	The participant wanted the return-navigation issue fixed and suggested making greater use of VR exploration by adding small discoverable surprises or easter-egg-style rewards to encourage users to look around the environment.

Summary Notes

Additional observations:

Most core interactions were understandable. The participant particularly liked the physical opening of doors and responded positively to the overall flow. The main interruptions occurred during Chapter selection and return navigation.

Main usability issues identified:

1. Incorrect Chapter labelling created unnecessary uncertainty.
2. The Navigation Panel was unavailable during the return workflow and required assistance.

Potential design improvements:

1. Check Chapter labels and Navigation Panel state before each test session.
2. Make the return-navigation workflow consistently available and reliable.
3. Consider optional VR-specific exploratory rewards or easter eggs after the core usability issues are resolved.

IP1 Testing Result – Participant P3

Participant ID	P3
Prototype	Interactive Prototype 1

Task Performance

Task	Success	Errors / Issues	Hesitation	Assistance	Key observation
------	---------	-----------------	------------	------------	-----------------

T1 Topic Selection	Yes	No major issue	No	Minimal	The participant understood the door as an entry point and was able to enter the AI podcast topic.
T2 Episode Browsing & Selection	Yes	No major issue	Low	No	The participant understood how to browse the available episodes and select an episode.
T3 Chapter Browsing & Selection	Yes, with explanation	The meaning of Chapter and labels such as Intro / AI Basics / Interview was unclear	Yes	Yes	The participant could use the chapter interface, but did not initially understand that chapters represented time-based sections within an episode.
T4 Favourite Interaction	Yes, with guidance	Favourite gesture was not immediately discoverable; Heart appeared too small and visually flat	Yes	Yes	The participant needed the favourite instruction repeated and was then told to drag the Heart. They suggested a larger, more physical 3D Heart.
T5 Global Navigation	Yes, with prompting	Return path was not fully self-explanatory	Yes	Yes	The participant required prompting during the return flow, indicating that navigation state and destination cues could be clearer.

Task-Specific Observations

Observation	Result
Recognised doors as content entry points	Yes
Understood Episode browsing and selection	Yes
Understood the meaning of Chapter without explanation	No
Understood Chapter as a time-based section after explanation	Yes
Completed the Favourite interaction without guidance	No
Preferred a larger, more physical 3D Heart	Yes
Overall Topic → Episode → Chapter → Experience flow felt clear	Yes
Navigation state and return cues were fully self-explanatory	No
Wanted more differentiation between content/door choices	Yes
Wanted richer visual content in the Podcast Experience	Yes

Post-Test Interview Notes

Question	Participant response / notes
----------	------------------------------

Q1. Which interaction was the easiest to understand, and why?	The participant found the door-based interaction easy to understand. Facing and entering a door clearly communicated how to access content, and the Episode selection process was also understandable.
Q2. Was there any point where the participant was unsure about what to do next?	The main uncertainty was the meaning of the Chapter options. Labels such as Intro, AI Basics and Interview did not initially communicate that they represented different time-based sections of the podcast episode.
Q3. What was the participant's initial interpretation of the Heart and Favourite Area?	The favourite interaction was not immediately clear from the visual design alone. The participant suggested making the Heart larger and more physical, using a 3D model rather than a flat 2D appearance, so that it looks more obviously interactive.
Q4. Did the Topic → Episode → Chapter → Podcast Experience flow feel clear? Why or why not?	The participant considered the overall flow clear once the content hierarchy was understood. However, clearer visual cues would help distinguish the Topic, Episode and Chapter levels and reduce uncertainty about the current location in the experience.
Q5. If one thing could be changed about the experience, what would the participant change?	The participant suggested enriching the Podcast Experience beyond subtitles, for example by presenting visual content on a television-style display. They also wanted clearer differentiation between content choices, such as more distinctive door appearances.

Summary Notes

Additional observations:

The participant responded positively to the overall structure and the use of doors as spatial content entry points. The core flow was understandable, but some labels and interaction affordances did not communicate their purpose clearly enough without explanation.

Main usability issues identified:

1. The Chapter concept is not self-explanatory; users need a clearer indication that chapters represent time-based sections of an episode.
2. The Heart favourite interaction lacks a strong physical affordance and requires clearer interaction guidance.
3. Topic, Episode and Chapter levels need stronger visual state cues to reduce navigation uncertainty.
4. Similar-looking doors/content entrances reduce quick visual recognition.
5. Subtitle-only content feels visually limited, and text readability could be improved.

Potential design improvements:

1. Add a short Chapter explanation or timeline cue near the Chapter list.
2. Increase the Heart size and replace the flat appearance with a clearly graspable 3D object.
3. Add clearer visual indicators for the current Topic / Episode / Chapter level.
4. Differentiate doors using distinct colours, textures or decorations.
5. Enrich the Podcast Experience with visual media, such as a television-style display, rather than relying only on subtitles.
6. Improve subtitle readability, including larger or adjustable text.

IP1 Testing Result – Participant P4

Participant ID	P4
Prototype	Interactive Prototype 1

Task Performance

Task	Success	Errors / Issues	Hesitation	Assistance	Key observation
T1 Topic Selection	Yes, with assistance	Basic movement control was unclear at the start	Yes	Yes	The participant did not initially know how to move in the environment and required a brief mouse-control prompt before successfully entering the AI Topic Door.
T2 Episode Browsing & Selection	Yes	No major issue observed	Low	No	The participant understood the Episode browsing interaction, viewed the available Episodes and selected an Episode.
T3 Chapter Browsing & Selection	Yes	No major issue observed	Low	Minimal	The participant browsed the available Chapters and successfully entered the Interview Chapter.
T4 Favourite Interaction	Yes, with assistance	Heart drag interaction was not self-explanatory	Yes	Yes	The participant recognised the Heart/Favourite Area as a favourite or collection feature, but did not know that the Heart needed to be dragged to complete the action.
T5 Global Navigation	Yes	Room transitions felt abrupt and spatial relationships were not strongly communicated	Low	Minimal	The participant completed the return flow and generally understood the hierarchy, but felt that moving between separate rooms lacked a clear spatial relationship.

Task-Specific Observations

Observation	Result
Understood basic movement without instruction	No
Recognised doors as content entry points	Yes
Understood Episode browsing and selection	Yes
Completed Chapter browsing and entry	Yes
Recognised Heart/Favourite Area purpose	Yes
Understood Heart drag interaction without guidance	No
Overall Topic → Episode → Chapter → Experience flow felt clear	Mostly yes
Felt room-to-room transitions were spatially meaningful	No
Preferred Navigation to communicate room relationships as a map	Yes
Wanted more freedom to move around the virtual rooms	Yes

Post-Test Interview Notes

Question	Participant response / notes
Q1. Which interaction was the easiest to understand, and why?	Entering rooms through doors was the easiest interaction to understand. The participant could immediately imagine going through a door to enter a different space.
Q2. Was there any point where the participant was unsure about what to do next?	The participant was initially unsure how to move in the environment. The Heart favourite interaction was also unclear because it was not obvious that the Heart needed to be dragged to the Favourite Area.
Q3. What was the participant's initial interpretation of the Heart and Favourite Area?	The participant recognised the area as a favourite or collection feature, but was unsure how to operate it and did not initially understand the drag interaction.
Q4. Did the Topic → Episode → Chapter → Podcast Experience flow feel clear? Why or why not?	The overall hierarchy was fairly clear, but moving suddenly between separate rooms felt abrupt. The participant suggested presenting Navigation as a spatial map rather than a flat list so that the relationship between the rooms is easier to understand.
Q5. If one thing could be changed about the experience, what would the participant change?	The participant wanted more freedom to move around and explore each room. The current experience felt too static and 2D-like for VR, so greater spatial movement and a map-based navigation concept would make the experience feel more immersive.

Summary Notes

Additional observations:

The door metaphor was the strongest interaction for this participant and supported the idea of entering different podcast spaces. The participant generally understood the content hierarchy, but wanted the experience to make stronger use of XR spatial movement and environmental exploration.

Main usability issues identified:

1. Basic movement controls are not introduced clearly at the start.
2. The Heart drag-to-favourite interaction requires guidance.
3. Room transitions feel abrupt and the spatial relationship between spaces is unclear.
4. The overall experience feels too static and does not yet provide enough freedom of movement for an XR environment.

Potential design improvements:

1. Add a short movement-control cue at the beginning of the experience.
2. Add stronger affordance or guidance for dragging the Heart to the Favourite Area.
3. Redesign Navigation as a spatial map that shows the relationship between rooms.
4. Allow more free movement and exploration within each virtual space to strengthen XR immersion.

IP1 Testing Result – Participant P5

Participant ID	P5
Prototype	Interactive Prototype 1

Task Performance

Task	Success	Errors / Issues	Hesitation	Assistance	Key observation
T1 Topic Selection	Yes, with minor assistance	Initial interaction required a brief control prompt	Low	Yes	The participant successfully entered the AI Topic Door after a short interaction prompt.
T2 Episode Browsing & Selection	Yes	Pagination requires stepping through Episodes one at a time	No	No	The participant browsed the available Episodes, selected Future of AI and continued successfully. They later suggested showing all Episode options simultaneously for direct selection.
T3 Chapter Browsing & Selection	Yes	No major issue observed	No	No	The participant browsed the Chapters and successfully selected and entered the Interview Chapter.
T4 Favourite Interaction	Yes, with assistance	Heart manipulation was not independently discoverable	Yes	Yes	The participant required guidance to move the Heart and complete the favourite action.
T5 Global Navigation	Yes	No major navigation issue observed	No	No	The participant independently noticed the Navigation control and successfully returned through the spatial hierarchy.

Task-Specific Observations

Observation	Result
Found button size and button interaction clear	Yes
Understood Topic → Episode → Chapter hierarchy	Yes
Completed Episode browsing and selection	Yes
Completed Chapter browsing and entry	Yes
Completed Heart favourite interaction without guidance	No
Independently discovered Navigation control	Yes
Understood podcast cover object without explanation	No
Wanted subtitle on/off control	Yes
Wanted all Episodes visible for direct selection	Yes
Considered the spatial door concept immersive	Yes
Felt the current environment was too static	Yes

Post-Test Interview Notes

Question	Participant response / notes
----------	------------------------------

Q1. Which interaction was the easiest to understand, and why?	The participant found the button-based controls easy to understand. The button size and interaction were clear, and the spatial door interaction also supported an immersive sense of entering the podcast.
Q2. Was there any point where the participant was unsure about what to do next?	The Heart favourite interaction required guidance because the method for manipulating the Heart was not obvious. The participant was also initially unsure about the purpose of the large podcast cover object until it was explained.
Q3. What was the participant's initial interpretation of the Heart and Favourite Area?	The participant recognised that the Heart related to the favourite function, but was unsure how to manipulate the Heart object to complete the interaction.
Q4. Did the Topic → Episode → Chapter → Podcast Experience flow feel clear? Why or why not?	Yes. The participant found the movement across the Topic, Episode, Chapter and Podcast Experience levels clear and understood the overall hierarchy.
Q5. If one thing could be changed about the experience, what would the participant change?	The participant suggested adding a subtitle on/off control, improving button placement, enriching the virtual environment to reduce the static feeling, and showing all Episodes at once so users can directly touch and select an option.

Summary Notes

Additional observations:

The participant responded positively to the button controls, content hierarchy and spatial door concept. The idea of opening doors and entering podcast spaces was considered immersive, although the current environment still felt visually static.

Main usability issues identified:

1. The Heart favourite interaction still requires guidance.
2. The purpose of the podcast cover object is not immediately clear.
3. The virtual environment is too static.
4. Subtitle controls and button placement could be improved.
5. Episode pagination makes the available choices less immediately visible.

Potential design improvements:

1. Improve the Heart interaction affordance so users can complete the favourite action independently.
2. Add a clear visual cue or label for the podcast cover object.
3. Add a subtitle on/off button and refine UI control placement.
4. Display all Episode options simultaneously for direct touch selection.
5. Enrich the virtual environment to strengthen immersion and reduce the static appearance.